Data Calibration Explanation

Data calibration ensures that numerical business metrics are **accurate, consistent, and reliable** before being used for analytics and reporting.

1. Why Data Calibration Is Required

In real-world data pipelines, transactional data may contain:

- Incorrect calculations
- Missing values
- Inconsistent numeric precision
- Business rule violations

Without calibration, such issues can lead to:

- Incorrect revenue reporting
- Misleading dashboards
- Poor business decisions

2. Calibration Scope

Data calibration is applied in the **Silver layer**, where data is cleaned and validated before aggregation.

The primary calibration focus is on **sales transaction metrics**.

3. Business Rule Definition

The following business rule is enforced:

total_amount = (quantity × unit_price) − discount

This rule defines the expected relationship between transaction attributes.

4. Calibration Logic Implementation

Step 1: Recalculate Total Amount

For each sales transaction, the expected total amount is recalculated using quantity, unit price, and discount.

Step 2: Validate Existing Values

- The recalculated value is compared with the source total_amount
- If the values do not match, the transaction is considered incorrect

Step 3: Correct Inconsistent Records

- Incorrect total_amount values are replaced with the recalculated value

- Correct values are retained as-is

This ensures consistent and accurate revenue metrics.

5. Handling Null and Invalid Values

- Null discounts are treated as zero where applicable
- Quantity and unit price are validated to be greater than zero
- Transactions failing validation rules are quarantined

6. Calibration and Quarantine Integration

Transactions that fail fundamental business rules (e.g., negative quantity or invalid references) are:

- Excluded from Silver clean data
- Stored in a quarantine table with rejection reasons

This ensures calibrated data is trustworthy while preserving rejected records for analysis.